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Crystal structure of E339K mutated human glucokinase reveals changes in the ATP binding site

Qiang Liu [a], Yunfeng Shen [b,c], Sanling Liu [a], Jianping Weng [b], Jinsong Liu [a,*]

[a] State Key Laboratory of Respiratory Diseases, Guangzhou Institutes of Biomedicine and Health, Chinese Academy of Sciences, Guangzhou 510530, China
[b] Department of Endocrinology, The Third Affiliated Hospital of Sun Yat-sen University, Research Center for Diabetes Care of Guangdong Province, Guangzhou 510630, China
[c] Department of Endocrinology and Metabolism, The Second Affiliated Hospital of Nanchang University, Nanchang 330006, China

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ABSTRACT

Human glucokinase (GK) plays an important role in glucose homeostasis. An E339K mutation in GK was recently found to be associated with hyperglycemia. It showed lower enzyme activity and impaired protein stability compared to the wild-type enzyme. Here, we present the crystal structure of E339K GK in complex with glucose. This mutation results in a conformational change of His416, spatially interfering with adenosine-triphosphate (ATP) binding. Furthermore, Ser411 at the ATP binding site is phosphorylated and then hydrogen bonded with Thr82, physically blocking the ATP binding. These findings provide structural basis for the reduced activity of this mutant.
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1. Introduction

Human glucokinase (GK) (adenosine-triphosphate (ATP): D-glucose 6-phosphotransferase, EC 2.7.1.2) is a 52 kD enzyme that can phosphorylate glucose to glucose-6-phosphate. It is mainly expressed in liver, pancreas, gut and brain [1,2]. Also known as hexokinase IV or D, GK is a member of the hexokinase family that catalyzes the first step in glycolysis, playing a crucial role in glucose homeostasis. However, GK exhibits sigmoidal kinetics instead of the Michaelis–Menten kinetics of non-allosteric hexokinases [3]. It is also different from non-allosteric hexokinases in its lack of product inhibition and relatively lower glucose affinity. In pancreatic β-cells, GK functions as a glucose sensor [4]. In liver, it regulates glucose uptake and storage [5]. Given its key role in glucose homeostasis, mutations of the *GK* gene have profound influence on its functions in human cells. To date, about 600 mutations have been reported for the *GK* gene, including non-sense, missense, and frameshift mutations [6,7]. A large number of GK mutations are inactivating mutations associated with maturity-onset diabetes of the young, type2 (MODY2). MODY2 is a form of diabetes mellitus caused by mutations in an autosomal dominant gene. A number of GK mutations, such as T51I, W99R, Y214C, V455M and A456V, are activating mutations that result in persistent hyperinsulinemic hypoglycemia of infancy (PHHI) [8–11].

In 2004, Katama and colleagues determined the crystal structures of human GK in both active and inactive forms [12]. Their results showed that GK has a small domain and a large domain separated by a deep cleft and undergoes a large conformational change by the rotation of the small domain induced by glucose binding. Their structure provides a general explanation of how the enzyme activity of GK can be affected by some of the mutations, especially those located in the glucose and ATP binding pockets and those involved in the conformation transition. Recently, a new inactivating mutation E339K was found by Shen et al. in a Chinese family with hyperglycemia [13]. This E339K mutation, however, is situated far from the ATP binding site. To understand how this mutation affects GK activity, we determined the crystal structure of E339K mutated GK. This first mutated GK crystal structure will shed light on how the mutation causes the enzyme kinetic alterations and will provide a better understanding on how the enzyme works.

2. Materials and methods

2.1. Protein expression and purification

The E339K GK, with an N-terminal MGHHHHHHENLYFQGM tag and corresponding amino acid residues 12–465, was cloned into

Abbreviations: GK, glucokinase; AMPPNP, adenylyl-imidodiphosphate; SEP, phosphorylated serine; ATP, adenosine-triphosphate; IPTG, isopropyl-β-D-thiogalactoside; DTT, dithiothreitol; TCEP, tris (2-carboxyethyl) phosphine; G6PDH, glucose-6-phosphate dehydrogenase; NADP+, nicotinamide adenine dinucleotide phosphate

* Corresponding author.
E-mail address: liu_jinsong@gibh.ac.cn (J. Liu).

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pET-21a using *Nde*I and *Xho*I cloning sites and verified by sequencing. *Escherichia coli* strain BL21 (DE3) was transformed with the constructed plasmid. The cells were cultured at 37 °C in 1 l of LB medium containing 100 μg/ml of ampicillin until OD_{600} reached a value of 0.7–1.0. The temperature was lowered to 20 °C and expression was induced by an addition of 0.1 mM isopropyl-β-D-thiogalactoside. After cultivating for 16 h, cells were harvested by centrifugation. All subsequent handling was at 4 °C. The pellets were resuspended in 50 mM Tris pH 8.0, 150 mM NaCl, 10% glycerol, 2 mM dithiothreitol (DTT), 50 mM imidazole and lysed by sonication. After centrifugation (30 min at 12,000×*g*), the supernatant was applied onto a Ni-NTA column (Qiagen). The target protein was eluted with 250 mM imidazole and loaded onto a Resource Q column (GE Healthcare) pre-equilibrated with 50 mM Tris pH 8.0, 150 mM NaCl, 10% glycerol and 2 mM DTT. Protein was eluted with a linear NaCl gradient (150–400 mM). Fractions containing GK were concentrated and loaded on a Superdex 200 column (GE Healthcare) pre-equilibrated with 20 mM Tris pH 8.0, 50 mM NaCl, and 5 mM tris (2-carboxyethyl) phosphine. Wild-type GK used in the following enzyme assay was prepared similarly.

2.2. Crystallization, data collection, and processing

For crystallization trials, glucose was added to the protein solution at a final concentration of 50 mM. Crystals of E339K GK were grown at 20 °C using the sitting drop vapor diffusion technique. The best crystals were obtained within two weeks by mixing equal volumes of protein solution (5 mg/ml) and reservoir solution containing 25% (w/v) PEG3350, 0.1 M HEPES pH 7.3, 0.2 M ammonium sulfate. For cryo data collection, crystals were sequentially immersed in mother liquors containing an additional 10% and 20% (v/v) glycerol as cryoprotectant. Data were collected at the Shanghai Synchrotron Radiation Facility, using beamline BL17U1 (λ = 0.9793 Å) at 100 K with a MX-225 CCD (Marresearch, Germany). The data were processed with MOSFLM [14] and SCALA from the CCP4 suite [15]. The structure was solved by molecular replacement using MOLREP in CCP4 and wild-type GK structure (PDB entry 1V4S) as the search model. Interactive model building and refinement were carried out using Coot [16] and PHENIX [17]. The final model was evaluated with MolProbity [18]. The coordinates and structure factors have been deposited in the Protein Data Bank under the accession code 3QIC. The data collection and structure refinement statistics are listed in Table 1. All structure figures were created using PyMOL [19].

2.3. Enzyme assay

Both wild-type and E339K mutant GK activities were measured using glucose-6-phosphate dehydrogenase (G6PDH)/nicotinamide adenine dinucleotide phosphate ($NADP^+$) coupled enzyme assay with some modification [20]. The assay buffer contained 50 mM HEPES pH 7.5, 50 mM KCl, 2 mM DTT, 0.1% BSA, 5 mM $MgCl_2$, 0.5 mM $NADP^+$ and 2.5 units/ml G6PDH. To determine glucose and ATP-related kinetic parameters, glucose was varied (0–100 mM) with a fixed saturating concentration of 5 mM ATP and then ATP was also varied (0–5 mM) with a fixed saturating concentration of 100 mM glucose. Assays were performed in a 96-well microplate with a final volume of 100 μL per well at 30 °C by monitoring the absorbance at 340 nm. The kinetic parameters were calculated as described previously [20,21].

2.4. Phosphorylation site analysis by mass spectrometry

Protein samples were prepared by SDS–PAGE and sent to Center of Technical Service for Life Sciences, University of Science and Technology of China, Hefei, China. Sample processing, data searching and analysis were all performed there.

Data-dependent MS/MS was performed. Electrospray spectra were acquired on a Thermo-Finnigan LTQ linear quadrupole ion trap mass spectrometer. The samples were digested by trypsin. Experimental results for the LTQ data were acquired using a C18 Reversed Phased Capillary Column. The mobile phase consisted of mobile A (0.1% formic acid in ddH_2O) and mobile B (100% acetonitrile). All spectra were obtained in positive ionization mode. Sample analyses were performed using data-dependent/dynamic exclusion with a repeat count of one to maximize coverage. All data files were searched against the human database.

Table 1
Data collection and refinement statistics

Data collection parameters	
Space group	$P4_32_12$
Cell dimensions (Å)	$a = b = 80.74$, $c = 178.41$
Resolution range (Å) (highest shell)	41.19–2.20 (2.32–2.20)
No. of observations	184 277
No. of unique reflections	30 831
Completeness (overall/outer shell) (%)	99.9/100.0
I/σ_I (overall/outer shell)	13.2/3.4
R_{merge} (%)	7.7(54.4)
Refinement parameters	
R_{work}/R_{free} (%)	21.3/25.2
No. of non-hydrogen atoms	3611
No. of water molecules	49
R.m.s deviations from ideal geometry	
Bond lengths (Å)	0.008
Bond angles (°)	1.097
Avg. B-factor	45.0
Ramachandran plot (%)	
Favored region	97.6
Allowed region	2.4

3. Results

3.1. Overall structure of E339K GK

The X-ray crystal structure of the E339K GK (glucose binding form) was determined at a resolution of 2.2 Å by molecular replacement. The crystal belongs to space group $P4_32_12$ with one molecule in the asymmetric unit. The refined structure of E339K GK consists of 453 residues, from Tyr7 to Lys459. The overall structure of the E339K GK is similar to the active form of wild-type GK (PDB entry 1V4S) with a root mean square deviation (r.m.s.d.) of 1.1 Å between Cα atoms of the two structures. Superimposing the mutant GK structure to the wild-type GK structure with only the large domain reveals a slight movement of the small domain (Fig. 1A). R.m.s.d. for the large and the small domain between the two structures is 1.0 and 1.6 Å, respectively. Significant differences are observed around the mutated residue.

The electron density map for K339 and surrounding residues is shown in Fig. 2A. The E339K mutation was also confirmed by mass spectrometry (data not shown). Surprisingly, the electron density map clearly showed extra density around the OG atom of Ser411 (Fig. 2B). It can be modeled very well as a phosphate (PO_3) covalently linked to residue Ser411. Mass spectrometry data confirmed the phosphorylation at Ser411. One phosphate was found in the peptide from residues 404–414 and was located in Ser411 (Fig. 3). No phosphorylation at Ser411 was found in the wild-type GK (data not shown).

3.2. Structural comparison of E339K with wild-type GK

In wild-type structure, Glu339 is located in the loop between helix α9 (residues 331–338) and helix α10 (residues 345–355).

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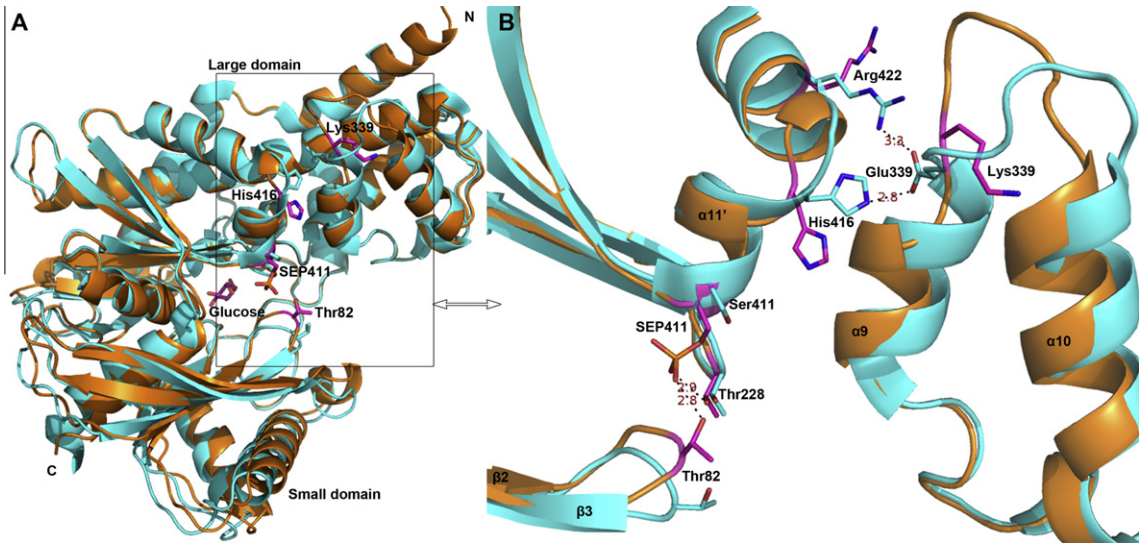


Fig. 1. Comparison of wild-type and E339K GK structures with superimposition using only the large domain. Wild-type GK complexed with glucose (PDB entry 1V4S, cyan). E339K GK complexed with glucose (orange). Hydrogen bonds shown as dashes, distances also indicated (Å). Residues with significant changes colored, 1V4S in cyan and E339K in magenta. (A) Overall comparison of wild-type and E339K GK. (B) Conformational variations around residue 339.

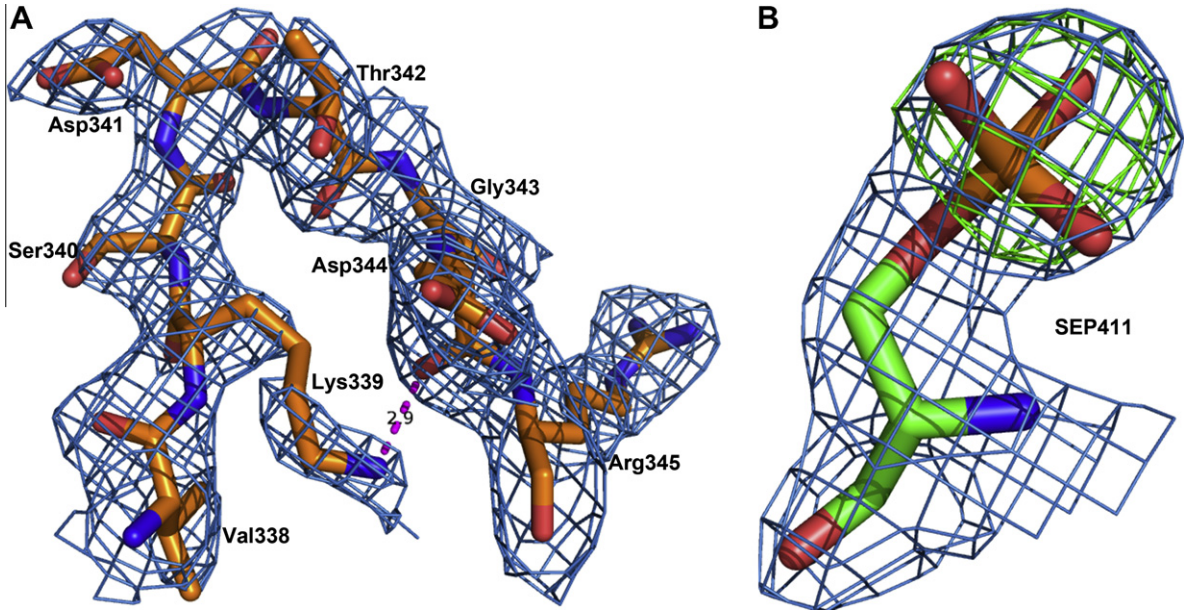


Fig. 2. The quality of electron density. (A) $2mF_o - DF_c$ map around mutated residue 339, contoured at 0.8σ. (B) Electron density at phosphorylated serine 411. $2mF_o - DF_c$ map (contoured at 1.5σ, blue), $mF_o - DF_c$ omit map (contoured at 5.0σ, green).

Several wild-type structures (PDB codes: 1V4S, 3ID8 and 3F9M) show that Glu339 forms two hydrogen bonds with His416 and Arg422 (Fig. 1B). The substituted Lys339 would produce unfavorable interactions with the residues mentioned above if it adopts the conformation as Glu339. Instead, Lys339 takes the opposite orientation and forms a hydrogen bond with the carbonyl oxygen of Asp344 (Fig. 2A). With the loss of hydrogen bonding, the side chain of His416 is rotated by nearly 90° and the side chain of Arg422 is more floppy in the mutant structure. Also, helix α9 is shorter as the mutation perturbs the helix formation.

Thr82 is located in the loop (residues 79–84) region linking two β-sheets (β2 and β3). In the mutant structure, the main chain of this loop moves notably closer to the large domain (Fig. 1B). The Cα atom of Thr82 shifts about 3 Å toward the short helix α11′ (residues 412–416). Ser411 is positioned at the N terminus of helix α11′. Phosphorylation of Ser411 provides a new interaction with Thr82 (Fig. 1B). The phosphate moiety of SEP411 also interacts with the short loop (residues 226–230) and forms a hydrogen bond with the hydroxyl group of Thr228.

4. Discussion

A novel mutation, E339K, was found in a Chinese family by Shen et al. [13]. According to clinical features and laboratory examinations, the patients were diagnosed with fasting hyperglycemia. The kinetics analysis showed lower affinity for glucose and ATP, resulting in the dramatically decreased relative activity of E339K [13]. To better understand the mechanism of how E339K mutation

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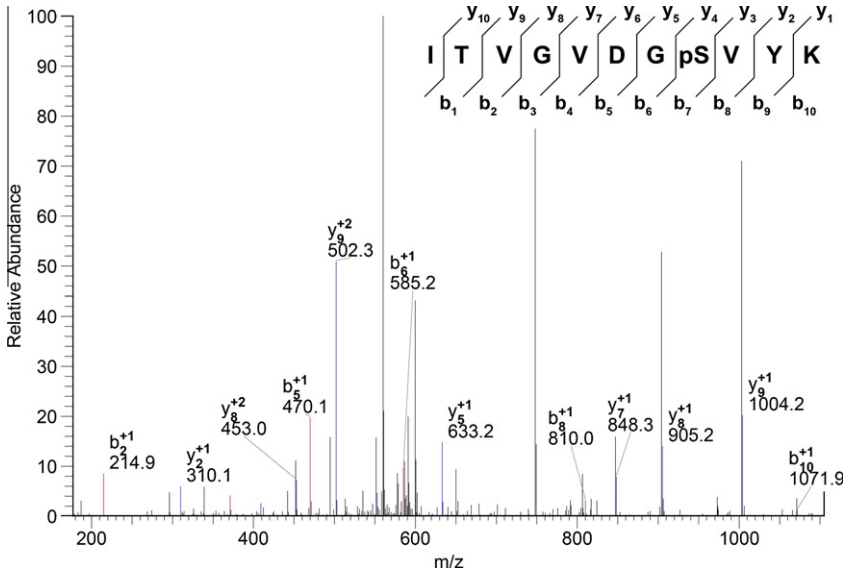


Fig. 3. Mass spectrometric analysis of E339K GK. MS/MS spectrum of E339K GK peptide at 1217 Da. This peptide, residues 404–414, contains a phosphorylation site. Phosphorylated amino acids are shown in the sequence. Peaks labeled using Bioworks Browser 3.3.

is associated with hyperglycemic phenotype, we determined the first GK mutant crystal structure and showed its structural differences with wild-type GK. Superimposing the crystal structure of wild-type GK complexed with glucose and the non-hydrolyzable ATP analog adenylyl-imidodiphosphate (AMPPNP) (PDB entry 3ID8, unpublished data) with our current structure (Fig. 4A), we can clearly understand why this mutation causes loss of enzymatic activity. The structures manifest conformational changes in the Glu339 loop region as a result of a single mutation. Glu339 establishes two hydrogen bonds with His416 and Arg422 (Fig. 1B). The hydrogen bond between Glu339 and His416 is the major stabilizing force for the correct conformation of His416 in the wild-type structure. In the mutant structure, the E339K disrupts the hydrogen bond between Glu339 and His416. Consequently, His416 adopts another conformation by changing the orientation of its side chain. Therefore, the imidazole ring of His416 would be situated in close vicinity of the AMPPNP adenine ring based on the superposition of the E339K structure and 3ID8 (Fig. 4A). The distance between His416 NE2 and AMPPNP C2 will then only be about 1.5 Å. Apparently, this close contact would hinder the binding of AMPPNP. The ATP binding will have to take another orientation in this situation.

We observed phosphorylated Ser411, confirmed by mass spectrometric analyses. A direct consequence of the phosphorylation at Ser411 is the formation of hydrogen bond interaction with Thr82, which is positioned in the loop (residues 79–84) linking β2 and β3. This hydrogen bond interaction pulls Thr82 closer to SEP411 and spatially blocks the entrance of the ATP binding pocket (Fig. 4A). From the surface view of the two structures (Fig. 4B and C), the ATP binding pocket has altered considerably in E339K, especially

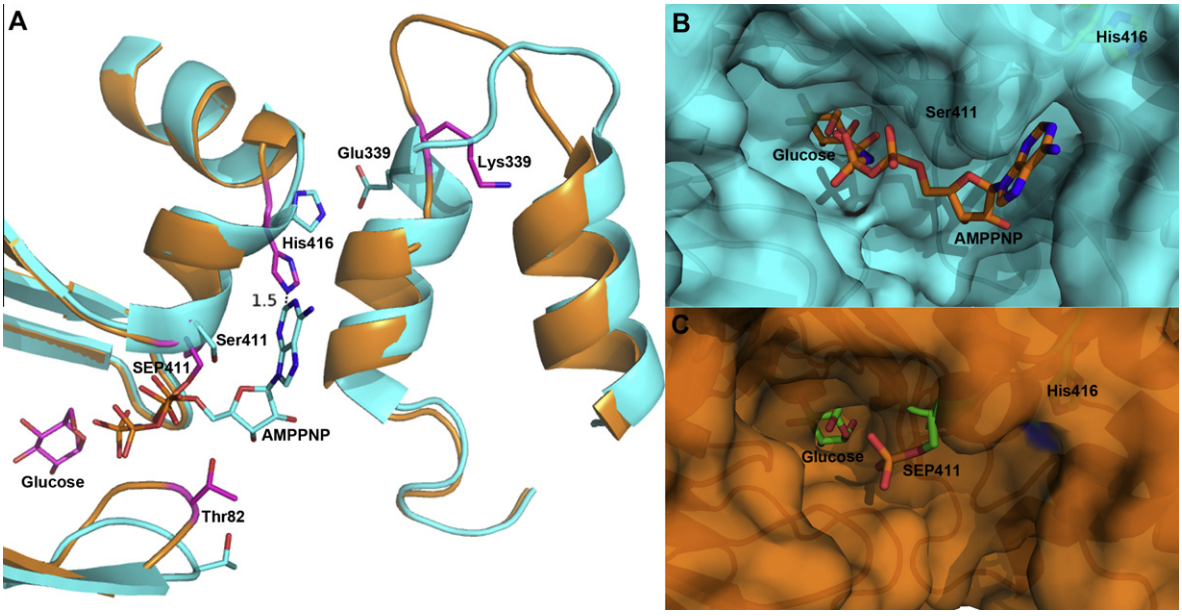


Fig. 4. Comparison of ATP binding pockets for wild-type and E339K GK. Wild-type GK complexed with glucose and AMPPNP (PDB entry 3ID8, cyan). E339K GK complexed with glucose (orange). (A) Diagram of wild-type and E339K GK. (B) Surface view of wild-type GK. (C) Surface view of E339 GK.

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Table 2
Kinetic constants of wild-type and E339K GK. The results from three preparations are recorded. Mean ± S.E. are given. The $S_{0.5}$ and nH (Hill coefficient) values were obtained from the Hill equation and K_m from the Michaelis–Menten equation [20]. GK phosphorylating relative activity was normalized relative to the wild-type value. Relative activity = (GK K_{cat}/glucose $S_{0.5}{}^{nH}$) (2.5/2.5 + ATP K_m) [21].

	Glucose $S_{0.5}$ (mmol/L)	nH	ATP K_m (mmol/L)	Relative activity
WT	8.65 ± 0.28	1.75 ± 0.09	0.32 ± 0.06	1.00
E339K	15.44 ± 0.48	1.60 ± 0.05	1.50 ± 0.18	0.20

at the residues 411 and 416. SEP411 and Thr82 evidently form a "closed door" and leave no space for ATP binding. This also corroborates well with the kinetics analysis (Table 2). For the E339K mutant, the glucose $S_{0.5}$ increased only two folds while the ATP K_m increased five folds comparing with the wild-type GK.

It has been reported before that serine residues in GK can be phosphorylated by protein kinase A in vitro [22]. For the first time, our structure showed that Ser411 at the ATP binding site can be phosphorylated and this phosphorylation has direct consequence on the ATP binding. Throughout the purification and crystallization processes, the protein was not exposed to any phosphate related compound. Therefore the formation of this phosphorylated residue should have occurred inside the cell. We speculate that this is an autophosphorylation utilizing the ATP in the *E. coli*. In the E339K structure, the original ATP binding pocket was blocked, thus the phosphate group that should have been added to the glucose was wrongly placed to the hydroxyl group of Ser411. Furthermore, Ser411 was completely phosphorylated with occupancy 1.0 from the electron density fitting (Fig. 2B). The strong presence of phosphorylated serine in the crystals implies that the phosphorylation should be stable enough against hydrolysis and persist for a considerably long period. But more data is needed to prove this hypothesis.

Up to date, several hundred mutations of GK have been found. But the relationship between the mutation and enzyme activity is unclear. In addition, there are few reports on the phosphorylation of GK. Here we present the first mutant crystal structure of GK and also report the first structural evidence of the phosphorylation of GK. Our structure reveals that the E339K mutation, along with the phosphorylation of Ser411 at the ATP binding site, induces the changes of the ATP binding pocket and results in loss of enzymatic activity. This provides structural insight into how the E339K mutation contributes to hyperglycemia. Further studies of Ser411 phosphorylation may add to our understanding of the mechanism of GK.

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References

[1] Matschinsky, F.M., Magnuson, M.A., Zelent, D., Jetton, T.L., Doliba, N., Han, Y., Taub, R. and Grimsby, J. (2006) The network of glucokinase-expressing cells in glucose homeostasis and the potential of glucokinase activators for diabetes therapy. Diabetes 55, 1–12.
[2] Efanov, A.M. et al. (2005) A novel glucokinase activator modulates pancreatic islet and hepatocyte function. Endocrinology 146, 3696–3701.
[3] Storer, A.C. and Cornish-Bowden, A. (1976) Kinetics of rat liver glucokinase. Co-operative interactions with glucose at physiologically significant concentrations. Biochem. J. 159, 7–14.
[4] Matschinsky, F.M., Glaser, B. and Magnuson, M.A. (1998) Pancreatic beta-cell glucokinase: closing the gap between theoretical concepts and experimental realities. Diabetes 47, 307–315.
[5] Postic, C. et al. (1999) Dual roles for glucokinase in glucose homeostasis as determined by liver and pancreatic beta cell-specific gene knock-outs using Cre recombinase. J. Biol. Chem. 274, 305–315.
[6] Gloyn, A.L. (2003) Glucokinase (GCK) mutations in hyper- and hypoglycemia: maturity-onset diabetes of the young, permanent neonatal diabetes, and hyperinsulinemia of infancy. Hum. Mutat. 22, 353–362.
[7] Osbak, K.K., Colclough, K., Saint-Martin, C., Beer, N.L., Bellanne-Chantelot, C., Ellard, S. and Gloyn, A.L. (2009) Update on mutations in glucokinase (GCK), which cause maturity-onset diabetes of the young, permanent neonatal diabetes, and hyperinsulinemic hypoglycemia. Hum. Mutat. 30, 1512–1526.
[8] Gloyn, A.L. et al. (2003) Insights into the biochemical and genetic basis of glucokinase activation from naturally occurring hypoglycemia mutations. Diabetes 52, 2433–2440.
[9] Cuesta-Munoz, A.L. et al. (2004) Severe persistent hyperinsulinemic hypoglycemia due to a de novo glucokinase mutation. Diabetes 53, 2164–2168.
[10] Glaser, B. et al. (1998) Familial hyperinsulinism caused by an activating glucokinase mutation. N. Engl. J. Med. 338, 226–230.
[11] Christesen, H.B. et al. (2002) The second activating glucokinase mutation (A456V): implications for glucose homeostasis and diabetes therapy. Diabetes 51, 1240–1246.
[12] Kamata, K., Mitsuya, M., Nishimura, T., Eiki, J. and Nagata, Y. (2004) Structural basis for allosteric regulation of the monomeric allosteric enzyme human glucokinase. Structure 12, 429–438.
[13] Shen, Y., Cai, M., Liang, H., Wang, H. and Weng, J. (2010). Insight into the biochemical characteristics of a novel glucokinase gene mutation. Hum. Genet. DOI 10.1007/s00439-010-0914-4.
[14] Leslie, A.G.W. (1992). Recent changes to the MOSFLM package for processing film and image plate data. Joint CCP4 + ESF-EAMCB Newsletter on Protein Crystallography 26.
[15] Collaborative Computational Project Number 4. (1994). The CCP4 suite: programs for protein crystallography (1994) Acta Crystallogr. D Biol. Crystallogr. 50, 760–763.
[16] Emsley, P. and Cowtan, K. (2004) Coot: model-building tools for molecular graphics. Acta Crystallogr. D Biol. Crystallogr. 60, 2126–2132.
[17] Adams, P.D. et al. (2010) PHENIX: a comprehensive python-based system for macromolecular structure solution. Acta Crystallogr. D Biol. Crystallogr. 66, 213–221.
[18] Chen, V.B. et al. (2010) MolProbity: all-atom structure validation for macromolecular crystallography. Acta Crystallogr. D Biol. Crystallogr. 66, 12–21.
[19] Schrodinger, LLC. (2010) The PyMOL Molecular Graphics System, Version 1.3r1.
[20] Kesavan, P., Wang, L., Davis, E., Cuesta, A., Sweet, I., Niswender, K., Magnuson, M.A. and Matschinsky, F.M. (1997) Structural instability of mutant beta-cell glucokinase: implications for the molecular pathogenesis of maturity-onset diabetes of the young (type-2). Biochem. J. 322 (Pt 1), 57–63.
[21] Pino, M.F. et al. (2007) Glucokinase thermolability and hepatic regulatory protein binding are essential factors for predicting the blood glucose phenotype of missense mutations. J. Biol. Chem. 282, 13906–13916.
[22] Ekman, P. and Nilsson, E. (1988) Phosphorylation of glucokinase from rat liver in vitro by protein kinase A with a concomitant decrease of its activity. Arch. Biochem. Biophys. 261, 275–282.

18733468, 2011, 8, Downloaded from https://febs.onlinelibrary.wiley.com/doi/10.1016/j.febslet.2011.03.026 by Spanish Cochrane National Provision (Ministerio de Sanidad), Wiley Online Library on [21/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License